

PROJECT TOPIC: WETLAND VISIT: MIGRATORY BIRDS & HABITAT STUDY

Topic Description: Field Study of a Freshwater/Coastal Wetland in Maharashtra
During October Arrival: Water Levels, Biodiversity, and Central Asian Flyway
Migratory Avifauna

Wetlands—encompassing freshwater marshes, intertidal mudflats, shallow lakes, backwaters, and reservoirs—are hailed globally as the 'kidneys of the landscape' due to their remarkable capacity to filter aquatic pollutants, buffer floodwaters, recharge groundwater, and support extraordinary biological diversity. In India, wetlands represent vital lifelines for millions of long-distance winter migratory waterbirds that travel thousands of kilometers along the **Central Asian Flyway (CAF)** to escape the freezing sub-zero Arctic winters of Siberia, Mongolia, and Central Asia.

This project documents an extensive field visit and ecological habitat study of a prominent **Freshwater / Coastal Wetland in Maharashtra** (surveying a representative site such as Bhigwan wetland on Ujani reservoir, Pashan wetland in Pune, or the Thane Creek Ramsar site) conducted during the crucial winter arrival month of **October**.

The study documents seasonal changes in wetland water surface levels, submerged and emergent flora, aquatic invertebrate food supplies, and the arrival of both resident and trans-continental long-distance migratory bird species. By analyzing pressing threats—such as municipal sewage inflow, plastic pollution, and siltation—the project outlines actionable community conservation strategies.

2. Importance of the topic (विषयाचे महत्त्व)

Wetland ecological field surveys provide indispensable insights into global wildlife flyways and regional hydrological health:

Why Wetland & Migratory Bird Studies Matter :-

- **International Flyway Interconnectedness:** Migratory birds like the Bar-headed Goose, Northern Pintail, and Greater Flamingo connect India to global biomes across Siberia and the Himalayas. Protecting local wintering wetlands in Maharashtra is essential to prevent international species extinction.
- **Living Indicators of Environmental Catchment Health:** Waterbirds sit at the apex of the aquatic food web. A sharp drop in migratory bird numbers serves as an immediate early warning of toxic pesticide runoff, bioaccumulation, or collapsing fish stocks in the catchment.
- **Essential Hydrological Sponge Functions:** Wetlands act as giant detention basins that absorb monsoon floodwaters, filter agricultural chemical pollutants through reed beds, and prevent coastal saline water intrusion.
- **Cultivating Global Environmental Perspectives:** Observing an Arctic Tern or Garganey that has flown 6,000 km transforms student thinking, connecting local neighborhood ponds to planetary ecological survival.

Wetlands prove that local water bodies are globally significant biological heritage sites.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The field study was guided by structured limnological, ornithological, and migratory tracking objectives:

Structured Project Objectives :-

- • **Wetland Hydrology & Water Level Recording:** To record post-monsoon water surface levels, shoreline mudflat exposure, and physical water clarity in the month of October.
- • **Floral & Invertebrate Survey:** To inventory dominant aquatic macrophytes (Hydrilla, Lotus, Typha) and benthic invertebrates providing essential food resources.
- • **Migratory vs. Resident Avian Census:** To observe, count, and classify bird species into local resident breeding waterbirds versus trans-continental winter migrants.
- • **Central Asian Flyway Route Mapping:** To document the origin countries, breeding grounds, and migration distances of key wintering waterfowl.
- • **Threat Audit & Community Conservation Plan:** To identify anthropogenic pressures—including sewage discharge, fishing net entanglements, and plastic debris—and formulate local mitigation measures.

The scope encompasses early morning point-count surveys, spotting scopes, and verified BNHS census literature.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized wetland bird census and habitat assessment methodology was deployed:

Methodological Survey Steps :-

- **Phase 1 – Field Site Selection & Timing:** Selected an accessible representative wetland site and conducted observation sessions in mid-October during peak morning bird activity (6:00 AM to 9:30 AM).
- **Phase 2 – Point Count & Line Transect Census:** Established three fixed shoreline vantage points, utilizing 10x50 binoculars and a 20-60x spotting scope to count and identify waterbird flocks without causing disturbance.
- **Phase 3 – Water Level & Macrophyte Assessment:** Recorded shoreline waterline marks, mudflat exposure width, and the presence of submerged aquatic plants (*Hydrilla*) and emergent reed beds (*Typha*).
- **Phase 4 – Flyway Data Correlation:** Researched the breeding ranges and migration routes of observed migratory birds using BNHS Asian Waterbird Census databases.
- **Phase 5 – Report Synthesis:** Compiled findings into structured matrices and aligned results with Chapter 7 EVS curriculum competencies.

All species were identified using standard field guides authored by Dr. Sálim Ali and Richard Grimmett.

5. Observations (निरीक्षणे)

Systematic observations recorded during the mid-October field survey revealed dynamic post-monsoon ecological conditions and arriving migratory flocks:

5.1 Tabular Record of Avian Biodiversity Observed in October

Common & Scientific Name	Status in Wetland	Breeding Origin & Flyway Route	Feeding Behavior & Ecological Role
Northern Pintail <i>Anas acuta</i>	Winter Migrant (Flock of ~65)	Siberia & Northern Europe; crosses Himalayas via Central Asian Flyway.	Dabbling duck; feeds in shallow water by tipping tail-up to consume submerged seeds, roots, and aquatic snails.
Garganey <i>Spatula querquedula</i>	Winter Migrant (Flock of ~40)	Temperate Eurasia and Central Asia; arrives early in September-October.	Small migratory duck skimming floating surface seeds and small aquatic crustaceans; roosts in open deep water.
Bar-headed Goose <i>Anser indicus</i>	Winter Migrant (Small group of 12)	High-altitude plateau lakes of Tibet and Mongolia; flies over Mt. Everest (8,000m+).	Grazes on shoreline agricultural winter wheat shoots and gram crops; unique hemoglobin adaptation for extreme altitudes.
Purple Swampphen <i>Porphyrio poliocephalus</i>	Resident Breeder (Abundant, 50+)	Native across Indian subcontinent wetlands; non-migratory.	Large rail with massive red bill and long toes; walks on floating vegetation; pulls tender Typha reed shoots.
Painted Stork <i>Mycteria leucocephala</i>	Local Migrant (Colony of 28)	Breeds in regional Indian heronries; moves locally following water depth.	Tactile feeder; wades through shallow muddy water with open bill immersed, snapping shut upon touching fish.

5.2 Hydrological Observations Recorded in October

6. Analysis of data (माहितीचे विश्लेषण)

Analytical examination of the wetland field data demonstrates the delicate ecological mechanics of bird migration and wetland health:

Core Analytical Inferences :-

- **The Crucial Importance of the October Arrival Window:** Migratory waterfowl arrive in Maharashtra exhausted after non-stop flights of 4,000 to 6,000 kilometers over the high Himalayas, having burned up to 50% of their body fat. The availability of high-energy submerged vegetation (*Hydrilla*) and aquatic snails in October is a matter of life or death for their physical survival.
- **Niche Partitioning Among Co-Existing Waterbirds:** Competition is minimized through distinct feeding niches: Pintails feed on the bottom in 30 cm water by tipping upside-down, Garganey skim surface seeds, Painted Storks wade in knee-deep water for fish, and Plovers forage exclusively on exposed muddy shores.
- **The Threat of Premature Water Drainage & Siltation:** In many multipurpose reservoirs, irrigation canals draw water too rapidly in early winter, draining shallow mudflats prematurely and leaving migratory waterfowl without feeding habitat by January.
- **Sewage Pollution & Toxic Blue-Green Algae:** Inflow of untreated municipal sewage causes toxic cyanobacterial blooms, which have been scientifically linked to fatal avian botulism outbreaks that killed thousands of migratory waterbirds at Sambhar Lake.

The analysis proves that protecting wetlands requires maintaining natural seasonal water level regimes.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Maharashtra's wetlands are indispensable global biodiversity hubs that sustain the awe-inspiring trans-continental migrations of the Central Asian Flyway. Conserving these ecosystems requires strict statutory protection under the Wetland Rules, complete sewage interception, and regulated agricultural water drawing.

Action Checklist for Wetland Conservation :-

- ❑ Advocate for the official notification of local wetlands under the National Wetland Rules, 2017.
- ❑ Participate in the annual Asian Waterbird Census (AWC) conducted every January by BNHS.
- ❑ Stop the dumping of single-use plastic bottles and post-immersion clay debris into wetland margins.
- ❑ Prohibit the installation of fine monofilament fishing gillnets that accidentally drown diving ducks and cormorants.
- ❑ Maintain wide buffer zones of native trees around wetlands to filter incoming agricultural pesticide runoff.

8. Bibliography / References (संदर्भ सूची)

1. Ali, Sálim. (2002). *The Book of Indian Birds (13th Edition)*. Oxford University Press & BNHS, Mumbai.
2. Ministry of Environment, Forest and Climate Change. (2017). *Wetlands (Conservation and Management) Rules*. New Delhi.
3. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.
4. Prasad, S. N., et al. (2002). *Conservation of Wetlands of India: A Review*. Salim Ali Centre for Ornithology (SACON).